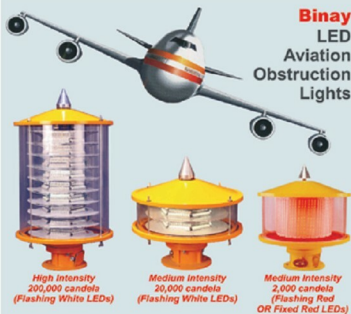


BINAY - Lighting Up The Future



BINAY LED-based HIGH, MEDIUM AND LOW Intensity Aviation Obstruction Light Beacons

As per International Civil Aviation Organisation (ICAO) requirements Available in Low Intensity, Medium Intensity and High Intensity versions (as per International Civil Aviation Organization guidelines), BINAY's patented LED Aviation Lights come with 5-year/3-year warranties

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick Installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED Intensity degradation over its operating lifetime



THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT

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Chi Hwan Lee, a mechanical engineering doctoral candidate, and the study's contributors in Scientific Reports.

While it was feasible for manufacturers to create solar cells on paper, textiles and rubber in the past, the strips were easily deformed in handling. By fabricating peel-and-stick solar cells, scientists have now made an attempt to broaden the uses of thin-film photovoltaic cells. Authors have also suggested use of solar cells with portable power supplies, electronic smart clothing and aerospace systems.

"Since the peel-and-stick process does not require any fabrication on the final target substrate, it circumvents all the fabrication challenges associated with these non-conventional substrates discussed above. Importantly, efficiency of transferred thin-film solar cells on any target substrate remains the same as that of thin-film solar cells fabricated on Si wafers," mentioned Lee and his fellow researchers in their study.

Ultra-thin electronics cooler

General Electric (GE) has launched what it claims to be the thinnest, high-performance cooler for ultra-thin tablets and laptops. Called dual piezoelectric cooling jets (DCJ), this cooler allows more space for other components. Interestingly, it uses just half the power of a comparable fan, enabling a significant boost to battery life. It is almost silent too.

According to Chris Giovannelli, vice president of GE, "DCJ can be made so quiet that users would not even know it is running."

"DCJ basically acts as a miniature pair of bellows: Expanding to suck in cool air, and then contracting to expel hot air. GE originally invented DCJ to help cool commercial jet engines, but two years ago it seems someone had the clever idea of miniaturising the tech for use in computers — and so here we are," ExtremeTech said in a report.

GE's cooler is roughly the size and thickness of a credit card. However, the complete cooling solution including a heatsink/pipe is 4 mm.

Hybrid solar-thermoelectric system that yields both electricity and heat

Researchers at MIT have come up with a low-cost system that can simultaneously produce electricity and hot water using the energy of the sun. Their design is based on a conventional solar thermal system but incorporates special features that make it more efficient and flexible.

For example, it uses evaporation, condensation and gravity to move the captured heat to the point of consumption — no need for an energy-consuming pump. It generates electricity using a solid-state technology that converts the captured solar energy directly into electricity with no moving parts. Depending on the materials used, the heat output can range from low-temperature water for household use to high-temperature process heat for industrial applications such as aluminium smelting. The researchers are now building a prototype of their system.

Evelyn Wang, associate professor of mechanical engineering and director of the MIT Device Research Laboratory,